Disciplina: **ENE0025 – Protocolos de Transporte e Roteamento**
Curso: **Engenharia de Redes de Comunicação**
Instituição: **Universidade de Brasília (UnB)**
Departamento: **Engenharia Elétrica**

Professor Responsável: **Prof. Dr. Laerte Peotta de Melo**

Relatório do Laboratório 5

Identificação

- Nome: **Artur Kohara Guerra**
- Matrícula: **231025181**
- Turma: **01**

Objetivo

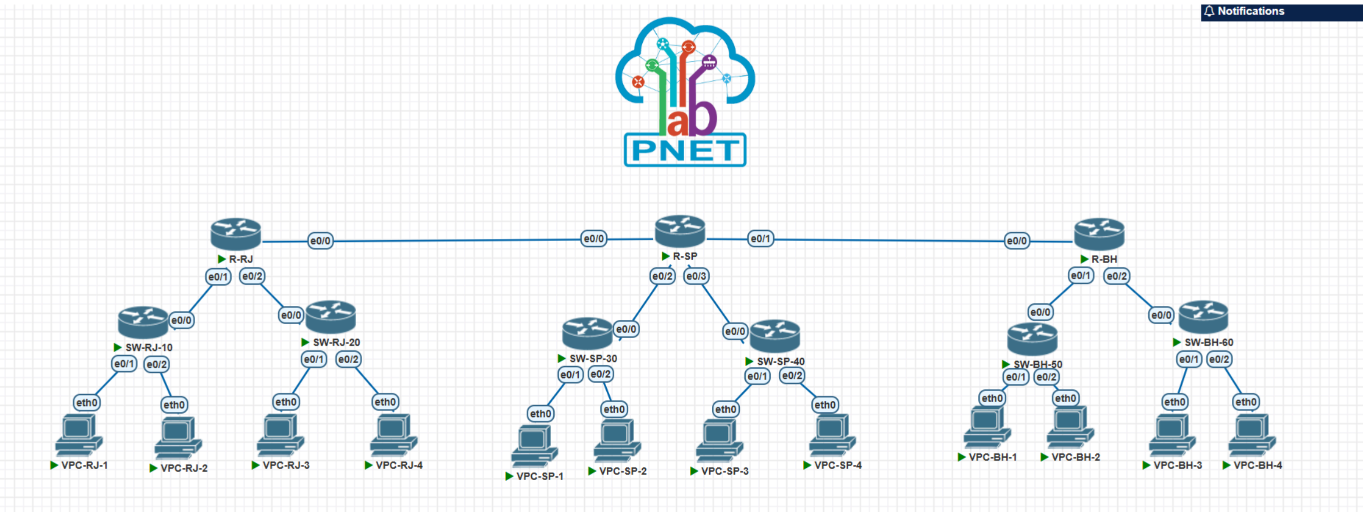
O objetivo deste experimento é configurar e validar o roteamento dinâmico em uma topologia composta por três roteadores, utilizando os protocolos **RIP** e **OSPF**, de forma a comparar seus comportamentos, características e desempenho em um mesmo cenário de rede.

Topologia do Laboratório

A topologia implementada no ambiente PNetLab consiste em:

- 3 roteadores interligando três unidades:
  - Router-RJ - Rio de Janeiro
  - Router-SP - São Paulo
  - Router-BH - Belo Horizonte
- 6 switches (2 por unidade)
- 12 hosts (4 por unidade)
- 2 enlaces WAN:
  - RJ ↔ SP
  - SP ↔ BH

Cada roteador conecta duas redes locais (LANs) e realiza o encaminhamento de pacotes entre as diferentes unidades.



Redes utilizadas

Segmento	Rede	Máscara
LAN RJ-10	172.16.10.0	255.255.255.0
LAN RJ-20	172.16.20.0	255.255.255.0
WAN RJ-SP	172.16.100.0	255.255.255.0
LAN SP-30	172.16.30.0	255.255.255.0
LAN SP-40	172.16.40.0	255.255.255.0
WAN SP-BH	172.16.200.0	255.255.255.0
LAN BH-50	172.16.50.0	255.255.255.0
LAN BH-60	172.16.60.0	255.255.255.0

Endereçamento das interfaces dos roteadores

Router-RJ

Interface	Endereço IP	Máscara	Função
`e0/0`	172.16.100.1	255.255.255.0	WAN para São Paulo
`e0/1`	172.16.10.254	255.255.255.0	LAN RJ-10
`e0/2`	172.16.20.254	255.255.255.0	LAN RJ-20

Router-SP

Interface	Endereço IP	Máscara	Função
`e0/0`	172.16.100.2	255.255.255.0	WAN para Rio de Janeiro

Interface	Endereço IP	Máscara	Função
`e0/1`	172.16.200.1	255.255.255.0	WAN para Belo Horizonte
`e0/2`	172.16.30.254	255.255.255.0	LAN SP-30
`e0/3`	172.16.40.254	255.255.255.0	LAN SP-40

Router-BH

Interface	Endereço IP	Máscara	Função
`e0/0`	172.16.200.2	255.255.255.0	WAN para São Paulo
`e0/1`	172.16.50.254	255.255.255.0	LAN BH-50
`e0/2`	172.16.60.254	255.255.255.0	LAN BH-60

Endereçamento dos hosts

Host	Endereço IP	Máscara	Gateway
VPC-RJ-1	172.16.10.1	255.255.255.0	172.16.10.254
VPC-RJ-2	172.16.10.2	255.255.255.0	172.16.10.254
VPC-RJ-3	172.16.20.1	255.255.255.0	172.16.20.254
VPC-RJ-4	172.16.20.2	255.255.255.0	172.16.20.254
VPC-SP-1	172.16.30.1	255.255.255.0	172.16.30.254
VPC-SP-2	172.16.30.2	255.255.255.0	172.16.30.254
VPC-SP-3	172.16.40.1	255.255.255.0	172.16.40.254
VPC-SP-4	172.16.40.2	255.255.255.0	172.16.40.254
VPC-BH-1	172.16.50.1	255.255.255.0	172.16.50.254
VPC-BH-2	172.16.50.2	255.255.255.0	172.16.50.254
VPC-BH-3	172.16.60.1	255.255.255.0	172.16.60.254
VPC-BH-4	172.16.60.2	255.255.255.0	172.16.60.254

Procedimento

O procedimento do laboratório consistiu em:

- Configurar as interfaces de todos os hosts e roteadores
- Configurar o RIP em todos os roteadores e verificar a comunicação
- Remover o RIP de todos os roteadores
- Configurar o OSPF em todos os roteadores e verificar a comunicação

---

Configuração básica das interfaces

VPCs

Todos os VPCs desse laboratório foram configurados de acordo com a tabela de endereçamento apresentada na parte **Endereçamento dos hosts** seguindo o seguinte formato:

```
Formato básico: ip <ip_address> <subnet_mask> <gateway>

ip 172.16.60.2 255.255.255.0 172.16.60.254
show ip
save
```

Router-RJ

```
Router> enable
Router# configure terminal
Router(config)# hostname Router-RJ
Router-RJ(config)# interface e0/0
Router-RJ(config-if)# ip address 172.16.100.1 255.255.255.0
Router-RJ(config-if)# no shutdown
Router-RJ(config-if)# interface e0/1
Router-RJ(config-if)# ip address 172.16.10.254 255.255.255.0
Router-RJ(config-if)# no shutdown
Router-RJ(config-if)# interface e0/2
Router-RJ(config-if)# ip address 172.16.20.254 255.255.255.0
Router-RJ(config-if)# no shutdown
Router-RJ(config-if)# end
```

Router-SP

```
Router> enable
Router# configure terminal
Router(config)# hostname Router-SP
Router-SP(config)# interface e0/0
Router-SP(config-if)# ip address 172.16.100.2 255.255.255.0
Router-SP(config-if)# no shutdown
Router-SP(config-if)# interface e0/1
Router-SP(config-if)# ip address 172.16.200.1 255.255.255.0
Router-SP(config-if)# no shutdown
Router-SP(config-if)# interface e0/2
Router-SP(config-if)# ip address 172.16.30.254 255.255.255.0
```

```
Router-SP(config-if)# no shutdown
Router-SP(config-if)# interface e0/3
Router-SP(config-if)# ip address 172.16.40.254 255.255.255.0
Router-SP(config-if)# no shutdown
Router-SP(config-if)# end
```

Router-BH

```
Router> enable
Router# configure terminal
Router(config)# hostname Router-BH
Router-BH(config)# interface e0/0
Router-BH(config-if)# ip address 172.16.200.2 255.255.255.0
Router-BH(config-if)# no shutdown
Router-BH(config-if)# interface e0/1
Router-BH(config-if)# ip address 172.16.50.254 255.255.255.0
Router-BH(config-if)# no shutdown
Router-BH(config-if)# interface e0/2
Router-BH(config-if)# ip address 172.16.60.254 255.255.255.0
Router-BH(config-if)# no shutdown
Router-BH(config-if)# end
```

Configuração do RIP

Router-RJ

```
Router-RJ> enable
Router-RJ# configure terminal
Router-RJ(config)# router rip
Router-RJ(config-router)# network 172.16.0.0
Router-RJ(config-router)# end
Router-RJ#
```

Router-SP

```
Router-SP> enable
Router-SP# configure terminal
Router-SP(config)# router rip
Router-SP(config-router)# network 172.16.0.0
```

```
Router-SP(config-router)# end
Router-SP#
```

Router-BH

```
Router-BH> enable
Router-BH# configure terminal
Router-BH(config)# router rip
Router-BH(config-router)# network 172.16.0.0
Router-BH(config-router)# end
Router-BH#
```

Verificação do RIP

`show ip route`

- Router-RJ

```
Router-RJ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 11 subnets, 2 masks
C        172.16.10.0/24 is directly connected, Ethernet0/1
L        172.16.10.254/32 is directly connected, Ethernet0/1
C        172.16.20.0/24 is directly connected, Ethernet0/2
L        172.16.20.254/32 is directly connected, Ethernet0/2
R        172.16.30.0/24 [120/1] via 172.16.100.2, 00:00:14, Ethernet0/0
R        172.16.40.0/24 [120/1] via 172.16.100.2, 00:00:14, Ethernet0/0
R        172.16.50.0/24 [120/2] via 172.16.100.2, 00:00:14, Ethernet0/0
R        172.16.60.0/24 [120/2] via 172.16.100.2, 00:00:14, Ethernet0/0
C        172.16.100.0/24 is directly connected, Ethernet0/0
L        172.16.100.1/32 is directly connected, Ethernet0/0
R        172.16.200.0/24 [120/1] via 172.16.100.2, 00:00:14, Ethernet0/0
```

- Router-SP

```
Router-SP#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 12 subnets, 2 masks
R        172.16.10.0/24 [120/1] via 172.16.100.1, 00:00:20, Ethernet0/0
R        172.16.20.0/24 [120/1] via 172.16.100.1, 00:00:20, Ethernet0/0
C        172.16.30.0/24 is directly connected, Ethernet0/2
L        172.16.30.254/32 is directly connected, Ethernet0/2
C        172.16.40.0/24 is directly connected, Ethernet0/3
L        172.16.40.254/32 is directly connected, Ethernet0/3
R        172.16.50.0/24 [120/1] via 172.16.200.2, 00:00:10, Ethernet0/1
R        172.16.60.0/24 [120/1] via 172.16.200.2, 00:00:10, Ethernet0/1
C        172.16.100.0/24 is directly connected, Ethernet0/0
L        172.16.100.2/32 is directly connected, Ethernet0/0
C        172.16.200.0/24 is directly connected, Ethernet0/1
L        172.16.200.1/32 is directly connected, Ethernet0/1
```

- Router-BH

```
Router-BH#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 11 subnets, 2 masks
R        172.16.10.0/24 [120/2] via 172.16.200.1, 00:00:20, Ethernet0/0
R        172.16.20.0/24 [120/2] via 172.16.200.1, 00:00:20, Ethernet0/0
R        172.16.30.0/24 [120/1] via 172.16.200.1, 00:00:20, Ethernet0/0
R        172.16.40.0/24 [120/1] via 172.16.200.1, 00:00:20, Ethernet0/0
C        172.16.50.0/24 is directly connected, Ethernet0/1
L        172.16.50.254/32 is directly connected, Ethernet0/1
C        172.16.60.0/24 is directly connected, Ethernet0/2
L        172.16.60.254/32 is directly connected, Ethernet0/2
R        172.16.100.0/24 [120/1] via 172.16.200.1, 00:00:20, Ethernet0/0
C        172.16.200.0/24 is directly connected, Ethernet0/0
L        172.16.200.2/32 is directly connected, Ethernet0/0
```

Teste de conexão

```
VPC-RJ-1> ping 172.16.30.1

84 bytes from 172.16.30.1 icmp_seq=1 ttl=62 time=2.665 ms
84 bytes from 172.16.30.1 icmp_seq=2 ttl=62 time=1.814 ms
84 bytes from 172.16.30.1 icmp_seq=3 ttl=62 time=1.842 ms
^C
VPC-RJ-1> ping 172.16.40.1

84 bytes from 172.16.40.1 icmp_seq=1 ttl=62 time=2.560 ms
84 bytes from 172.16.40.1 icmp_seq=2 ttl=62 time=1.602 ms
84 bytes from 172.16.40.1 icmp_seq=3 ttl=62 time=1.477 ms
84 bytes from 172.16.40.1 icmp_seq=4 ttl=62 time=1.570 ms
^C
VPC-RJ-1> ping 172.16.50.1

84 bytes from 172.16.50.1 icmp_seq=1 ttl=61 time=2.717 ms
84 bytes from 172.16.50.1 icmp_seq=2 ttl=61 time=1.637 ms
84 bytes from 172.16.50.1 icmp_seq=3 ttl=61 time=1.735 ms
^C
VPC-RJ-1> ping 172.16.60.1

84 bytes from 172.16.60.1 icmp_seq=1 ttl=61 time=2.595 ms
84 bytes from 172.16.60.1 icmp_seq=2 ttl=61 time=2.069 ms
84 bytes from 172.16.60.1 icmp_seq=3 ttl=61 time=1.792 ms
84 bytes from 172.16.60.1 icmp_seq=4 ttl=61 time=1.777 ms
```

Remoção do RIP

Em cada roteador:

```
configure terminal
no router rip
end
write memory
```

Configuração do OSPF

Neste laboratório será utilizada apenas a **área 0**, com objetivo introdutório.

Router-RJ

```
Router-RJ> enable
Router-RJ# configure terminal
Router-RJ(config)# router ospf 64
```

```
Router-RJ(config-router)# network 172.16.10.0 0.0.0.255 area 0
Router-RJ(config-router)# network 172.16.20.0 0.0.0.255 area 0
Router-RJ(config-router)# network 172.16.100.0 0.0.0.255 area 0
Router-RJ(config-router)# end
```

Router-SP

```
Router-SP> enable
Router-SP# configure terminal
Router-SP(config)# router ospf 65
Router-SP(config-router)# network 172.16.30.0 0.0.0.255 area 0
Router-SP(config-router)# network 172.16.40.0 0.0.0.255 area 0
Router-SP(config-router)# network 172.16.100.0 0.0.0.255 area 0
Router-SP(config-router)# network 172.16.200.0 0.0.0.255 area 0
Router-SP(config-router)# end
```

Router-BH

```
Router-BH> enable
Router-BH# configure terminal
Router-BH(config)# router ospf 66
Router-BH(config-router)# network 172.16.50.0 0.0.0.255 area 0
Router-BH(config-router)# network 172.16.60.0 0.0.0.255 area 0
Router-BH(config-router)# network 172.16.200.0 0.0.0.255 area 0
Router-BH(config-router)# end
```

Verificação do OSPF

`show ip ospf neighbor`

- Router-RJ

```
Router-RJ#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.200.1      1   FULL/DR         00:00:31    172.16.100.2    Ethernet0/0
```

- Router-SP

```
Router-SP#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.200.2      1   FULL/DR         00:00:37    172.16.200.2    Ethernet0/1
172.16.100.1      1   FULL/BDR        00:00:32    172.16.100.1    Ethernet0/0
```

- Router-BH

```
Router-BH#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.200.1      1   FULL/BDR        00:00:32    172.16.200.1    Ethernet0/0
```

`show ip route`

- Router-RJ

```
Router-RJ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 11 subnets, 2 masks
C        172.16.10.0/24 is directly connected, Ethernet0/1
L        172.16.10.254/32 is directly connected, Ethernet0/1
C        172.16.20.0/24 is directly connected, Ethernet0/2
L        172.16.20.254/32 is directly connected, Ethernet0/2
O        172.16.30.0/24 [110/20] via 172.16.100.2, 00:00:40, Ethernet0/0
O        172.16.40.0/24 [110/20] via 172.16.100.2, 00:00:40, Ethernet0/0
O        172.16.50.0/24 [110/30] via 172.16.100.2, 00:00:30, Ethernet0/0
O        172.16.60.0/24 [110/30] via 172.16.100.2, 00:00:30, Ethernet0/0
C        172.16.100.0/24 is directly connected, Ethernet0/0
L        172.16.100.1/32 is directly connected, Ethernet0/0
O        172.16.200.0/24 [110/20] via 172.16.100.2, 00:00:40, Ethernet0/0
```

- Router-SP

```
Router-SP#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 12 subnets, 2 masks
O        172.16.10.0/24 [110/20] via 172.16.100.1, 00:00:34, Ethernet0/0
O        172.16.20.0/24 [110/20] via 172.16.100.1, 00:00:34, Ethernet0/0
C        172.16.30.0/24 is directly connected, Ethernet0/2
L        172.16.30.254/32 is directly connected, Ethernet0/2
C        172.16.40.0/24 is directly connected, Ethernet0/3
L        172.16.40.254/32 is directly connected, Ethernet0/3
O        172.16.50.0/24 [110/20] via 172.16.200.2, 00:00:24, Ethernet0/1
O        172.16.60.0/24 [110/20] via 172.16.200.2, 00:00:24, Ethernet0/1
C        172.16.100.0/24 is directly connected, Ethernet0/0
L        172.16.100.2/32 is directly connected, Ethernet0/0
C        172.16.200.0/24 is directly connected, Ethernet0/1
L        172.16.200.1/32 is directly connected, Ethernet0/1
```

- Router-BH

```
Router-BH#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      172.16.0.0/16 is variably subnetted, 11 subnets, 2 masks
O        172.16.10.0/24 [110/30] via 172.16.200.1, 00:01:01, Ethernet0/0
O        172.16.20.0/24 [110/30] via 172.16.200.1, 00:01:01, Ethernet0/0
O        172.16.30.0/24 [110/20] via 172.16.200.1, 00:01:01, Ethernet0/0
O        172.16.40.0/24 [110/20] via 172.16.200.1, 00:01:01, Ethernet0/0
C        172.16.50.0/24 is directly connected, Ethernet0/1
L        172.16.50.254/32 is directly connected, Ethernet0/1
C        172.16.60.0/24 is directly connected, Ethernet0/2
L        172.16.60.254/32 is directly connected, Ethernet0/2
O        172.16.100.0/24 [110/20] via 172.16.200.1, 00:01:01, Ethernet0/0
C        172.16.200.0/24 is directly connected, Ethernet0/0
L        172.16.200.2/32 is directly connected, Ethernet0/0
```

Teste de conexão

```
VPC-RJ-1> ping 172.16.30.1

84 bytes from 172.16.30.1 icmp_seq=1 ttl=62 time=3.473 ms
84 bytes from 172.16.30.1 icmp_seq=2 ttl=62 time=2.794 ms
84 bytes from 172.16.30.1 icmp_seq=3 ttl=62 time=2.500 ms
^C
VPC-RJ-1> ping 172.16.40.1

84 bytes from 172.16.40.1 icmp_seq=1 ttl=62 time=3.007 ms
84 bytes from 172.16.40.1 icmp_seq=2 ttl=62 time=1.906 ms
84 bytes from 172.16.40.1 icmp_seq=3 ttl=62 time=2.400 ms
^C
VPC-RJ-1> ping 172.16.50.1

84 bytes from 172.16.50.1 icmp_seq=1 ttl=61 time=3.328 ms
84 bytes from 172.16.50.1 icmp_seq=2 ttl=61 time=2.765 ms
84 bytes from 172.16.50.1 icmp_seq=3 ttl=61 time=2.951 ms
84 bytes from 172.16.50.1 icmp_seq=4 ttl=61 time=2.984 ms
^C
VPC-RJ-1> ping 172.16.60.1

84 bytes from 172.16.60.1 icmp_seq=1 ttl=61 time=3.238 ms
84 bytes from 172.16.60.1 icmp_seq=2 ttl=61 time=2.266 ms
84 bytes from 172.16.60.1 icmp_seq=3 ttl=61 time=2.343 ms
84 bytes from 172.16.60.1 icmp_seq=4 ttl=61 time=2.286 ms
84 bytes from 172.16.60.1 icmp_seq=5 ttl=61 time=1.914 ms
```

Comparação entre RIP e OSPF

- **1. Qual protocolo foi mais simples de configurar?**

  O protocolo RIP foi mais simples de configurar, pois sua configuração exige poucos comandos e utiliza apenas o anúncio da rede principal através do comando `network`.

- **2. Qual protocolo apresentou maior riqueza de informações operacionais?**

  O OSPF apresentou maior riqueza de informações operacionais, pois fornece detalhes sobre vizinhança, adjacência e estados da rede, por meio do comando `show ip ospf neighbor`.

- **3. Qual a principal métrica do RIP?**

  A principal métrica do RIP é a contagem de saltos (hop count). O protocolo considera melhor a rota que possui menor número de roteadores intermediários até o destino.

- **4. Qual algoritmo é usado pelo OSPF?**

  O OSPF utiliza o algoritmo SPF (Shortest Path First), também conhecido como algoritmo de Dijkstra, para calcular o melhor caminho entre as redes.

- **5. Qual protocolo tende a escalar melhor?**

O OSPF tende a escalar melhor.

- **6. Qual protocolo converge melhor em cenários maiores?**

  O OSPF apresenta melhor convergência em cenários maiores, pois atualiza rapidamente as informações de rota quando ocorre alguma mudança na rede, reduzindo o tempo de indisponibilidade e tornando a comunicação mais estável.